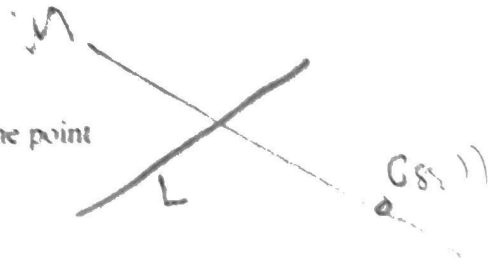


20 The straight line L has equation $y = 4x - 7$

The straight line M is perpendicular to L and passes through the point with coordinates (8, 1)

Find an equation for M

Give your answer in the form $y = mx + c$



$$m_1 \times m_2 = -1$$

$$4 \times m_2 = -1$$



$$m = -\frac{1}{4} = -\frac{1}{4}$$

$$M \rightarrow y = -\frac{1}{4}x + c$$

$\therefore$ substitute (8, 1) to find c

$$1 = -\frac{1}{4} \times 8 + c$$

$$1 = -2 + c$$

$$c = 1 + 2 = 3$$

$$\therefore y = -\frac{1}{4}x + 3$$

$$y = -\frac{1}{4}x + 3$$

(Total for Question 20 is 3 marks)

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